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10 / 10 submitted

Q1. Find the bugs, include the missing lines in the following snippet.

Code

```
#include <stdio.h>
```

```
int linearSearch(int arr[], int n, int key) {  
    for(int i = 0; i < n; i++) {  
        if(arr[i] == key)  
            return i;  
    }  
    return -1;  
}
```

```
int main() {  
    int arr[] = {10, 20, 30, 40, 50};  
    int n = 5;  
    int key = 30;  
  
    int result = linearSearch(arr, n, key);  
  
    if(result != -1)  
        printf("Element found at index %d\n", result);  
    else  
        printf("Element not found\n");  
  
    return 0;  
}
```

Input

5

10 20 30 40 50

30

Output

Q2. Buggy Code in an array

Code

```
#include <stdio.h>

int main() {
    int arr[5] = {10, 20, 30, 40, 50};

    for (int i = 0; i < 5; i++) {
        printf("%d ", arr[i]);
    }

    return 0;
}
```

Output

10 20 30 40 50

Q3. Identify an errors in the following snippet

Code

```
#include <stdio.h>

int binarySearch(int arr[], int n, int key)
{
    int low = 0, high = n - 1;
    while (low <= high)
    {
        int mid = (low + high) / 2;

        if (arr[mid] == key)
            return mid;
        else if (arr[mid] < key)
            low = mid + 1;
    }
}
```

```

        else
            high = mid - 1;
    }
    return -1;
}

int main()
{
    int n, key;
    if (scanf("%d", &n) != 1) return 0;
    if (scanf("%d", &key) != 1) return 0;

    int arr[n];
    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    printf("%d", binarySearch(arr, n, key));
    return 0;
}

```

Input

5

7

1 3 5 7 9

Output

3

Q4. Buggy Code in Stack Push Operation

Code

```
#include <stdio.h>
```

```
#define MAX 5
```

```
int stack[MAX];
```

```
int top = -1;
```

```
void push(int data) {  
    if (top == MAX - 1) {  
        printf("Stack Overflow");  
        return;  
    }  
    stack[++top] = data;  
}
```

```
int main() {  
    push(10);  
    push(20);  
  
    printf("%d", stack[top]);  
    return 0;  
}
```

Output

20

Q5. Buggy Code in Stack POP Operation

Code

```
#include <stdio.h>
```

```
#define MAX 5
```

```
int stack[MAX];
```

```
int top = 2;
```

```
int pop() {  
    if (top == -1)  
        return -1;  
    return stack[top--];  
}
```

```
}
```

```
int main() {  
    stack[0] = 10;  
    stack[1] = 20;  
    stack[2] = 30;  
  
    printf("%d\n", pop());  
    printf("%d\n", pop());  
    printf("%d\n", pop());  
    printf("%d\n", pop());  
    return 0;  
}
```

Output

30

20

10

-1

Q6. Buggy Code in underflow operation

Code

```
#include <stdio.h>
```

```
#define MAX 100
```

```
int stack[MAX];
```

```
int top = -1;
```

```
int pop() {  
    if (top == -1) {  
        printf("Underflow");  
        return -1;  
    }  
}
```

```
    return stack[top--];  
}
```

```
int main() {  
    printf("%d\n", pop());  
    return 0;  
}
```

Output

Underflow-1

Q7. Find the issue in the following code snippet.

Code

```
#include<stdio.h>  
  
#define MAX 5  
  
int queue[MAX];  
int front=0,rear=0;  
void enqueue(int x)  
{  
    if(rear==MAX)  
    {  
        printf("Overflow");  
        return;  
    }  
    queue[rear++]=x;  
    printf("Value is %d", queue[rear-1]);  
}  
  
int main()  
{  
    int x;  
    scanf("%d",&x);  
    enqueue(x);  
    return 0;
```

```
}
```

Input

25

Output

Q8. Buggy code in node creation

Code

```
#include<stdio.h>
#include<stdlib.h>

struct Node
{
    int data;
    struct Node *next;
};

int main()
{
    struct Node *newNode;
    newNode=(struct Node *)malloc(sizeof(struct Node));
    if(newNode==NULL)
    {
        printf("Memory Allocation failed");
        return 0;
    }
    newNode->data=10;
    newNode->next=NULL;
    printf("%d", newNode->data);
    free(newNode);
    return 0;
}
```

Output

Q9. Include missing statements and display the list of elements in linkedlist

Code

```

#include<stdio.h>

#include<stdlib.h>

struct Node
{
    int data;
    struct Node *next;
};

void display(struct Node *head)
{
    while(head!=NULL)
    {
        printf("%d->",head->data);
        head=head->next;
    }
    printf("NULL");
}

int main()
{
    struct Node n1,n2,n3;
    n1.data=10;
    n2.data=20;
    n3.data=30;
    n1.next=&n2;
    n2.next=&n3;
    n3.next=NULL;
    display(&n1);
    return 0;
}

```

Input

10->20->30->NULL

Output

Q10. 0Buggy code in node creation

Code

```
#include<stdio.h>

#include<stdlib.h>

> struct Node
{
    int data;
    struct Node *next;
};

int main()
{
    struct Node *temp;
    temp=(struct Node *)malloc(sizeof(struct
Node)); temp->data=25;
    temp->next=NULL;
    printf("%d",temp->data);
    free(temp);
    return 0;
}
```

Output

25